

TEACHING STATEMENT

by LILI JU

I chose Mathematics as the major during my undergraduate study (1991-1995) in Wuhan University in China partially because I had been deeply attracted by mathematics since I was very young. Since then I have met many many great mathematics teachers and professors later in my life, all in their own way inspired me with their love for the profession and taught me the patience and joy of sharing knowledge. From them I also learned to appreciate teaching and value the hard work it requires.

Regular Teaching Experiences

I went to the Iowa State University to pursue a Ph.D. degree in August 1998 after obtaining a M.S. Degree in Computational Mathematics from the Institute of Computational Mathematics in Chinese Academy of Sciences. During my first-year graduate study at Iowa State, I was a Teaching Assistant working in the mathematics help room, and responsible for answering problems from students on all introductory-level mathematical courses. From January 2001 to December 2001, I worked as a Teaching Assistant for two elementary undergraduate courses – *College Algebra* and *Calculus I*. My duties ranged from conducting recitations, to arranging and grading homework, quizzes and exams, to holding office hours for answering students' questions.

In August 2004, I joined the University of South Carolina (USC) at Columbia as an Assistant Professor of Mathematics after finishing a two-year's Industrial Postdoctoral training in the Institute for Mathematics and Its Applications (IMA) at the University of Minnesota. I was then promoted to Associate Professor with Tenure in August 2008 and to Full Professor in January 2013. I have taught various regular courses at USC for both undergraduate and graduate students. At the undergraduate level, I taught 11 different courses: *Calculus I&II* (MATH 141&142), *Vector Calculus* (MATH 241), *Elementary Differential Equations* (MATH 242), *Applied Linear Algebra* (MATH 344) *Financial Mathematics I&II* (MATH 514 & 515), *Ordinary Differential Equations* (MATH 520), *Numerical Linear Algebra* (MATH 526), *Numerical Analysis* (MATH 527), *Vector Analysis* (MATH 550), and *Analysis I* (MATH 554); at the graduate level, I taught 8 different courses: *Numerical Linear Algebra* (MATH 706), *Foundations of Computational Mathematics I&II* (MATH 708&709), *Differential Equations I&II* (MATH 723&724), *Numerical Analysis I&II* (MATH 726&727), *Numerical Differential Equations II* (Math 727) and *Scientific Computing I* (MATH 728J). I enjoy teaching elementary classes in all areas of mathematics for undergraduate students and courses in computational and applied mathematics (my research field) at the graduate level. I also had undergraduate and graduate student research credit courses such as Directed Readings & Research (MATH 798), Thesis Research (MATH 799), Dissertation Research (MATH 899), and Senior Thesis/Project (SCHC 499A).

For details of my teaching experiences and corresponding student evaluations at University of South Carolina, see the table below. In addition, I am currently serving as the course coordinator for Calculus II in Mathematics Department; I manage the Gateway Tests across all sections of the course, monitor students' progress and coordinate with instructors on providing advices and help to students in need to improve their performance.

TERM	NUMBER	COURSE TITLE	CRS.	ENROLL.	INSTR. EVAL.
Spring 2026	MATH 550	Vector Analysis	3	12	–
Fall 2025	MATH 344	Applied Linear Algebra	3	40	4.08/5
	MATH 514	Financial Mathematics I	3	22	4.10/5
Spring 2025	MATH 727	Numerical Differential Equations II	3	17	5.00/5
Spring 2024	MATH 242	Elementary Differential Equations	3	36	4.10/5
	MATH 344	Applied Linear Algebra	3	43	4.13/5

Fall 2023	MATH 142	Calculus II	4	56	4.18/5
	MATH 514	Financial Mathematics I	3	17	4.44/5
Spring 2023	MATH 727	Numerical Differential Equations II	3	5	5.00/5
Fall 2022	MATH 142	Calculus II	4	55	3.95/5
	MATH 554	Analysis I	3	15	4.46/5
Spring 2022	MATH 242	Elementary Differential Equations	3	36	4.10/5
Fall 2021	MATH 142	Calculus II	4	50	4.22/5
	MATH 514	Financial Mathematics I	3	18	3.85/5
Spring 2021	MATH 550	Vector Analysis	3	8	4.75/5
Fall 2020	MATH 142	Calculus II	4	108	4.32/5
Spring 2020	MATH 242	Elementary Differential Equations	3	43	4.24/5
	MATH 550	Vector Analysis	3	25	4.18/5
Fall 2019	MATH 142	Calculus II	4	47	3.75/5
	MATH 514	Financial Mathematics I	3	17	4.36/5
Spring 2019	MATH 242	Elementary Differential Equations	3	39	4.59/5
	MATH 344	Applied Linear Algebra	3	38	4.63/5
Fall 2018	MATH 241	Vector Calculus	3	43	4.03/5
	MATH 344	Applied Linear Algebra	3	50	4.10/5
Spring 2017	MATH 724	Differential Equations II	3	5	5.00/5
Fall 2016	MATH 723	Differential Equations I	3	7	4.71/5
Spring 2016	MATH 242	Elementary Differential Equations	3	39	4.48/5
	MATH 526	Numerical Linear Algebra	4	20	4.76/5
Fall 2015	MATH 520	Ordinary Differential Equations	3	35	4.44/5
Spring 2015	MATH 242	Elementary Differential Equations	3	40	4.86/5
	MATH 724	Differential Equations II	3	9	4.89/5
Fall 2014	MATH 241	Vector Calculus	3	39	4.57/5
	MATH 723	Differential Equations I	3	9	4.63/5
Spring 2014	MATH 142	Calculus II	4	61	4.24/5
	MATH 515	Financial Mathematics II	3	7	4.80/5
Fall 2013	MATH 514	Financial Mathematics I	3	17	3.67/5
Spring 2013	MATH 242	Elementary Differential Equations	3	36	4.61/5
	MATH 709	Foundations of Comput. Math. II	3	3	3.50/5
Fall 2012	MATH 241	Vector Calculus	3	45	4.47/5
	MATH 708	Foundations of Comput. Math. I	3	4	4.33/5
Spring 2012	MATH 142	Calculus II	4	63	4.24/5
	MATH 526	Numerical Linear Algebra	4	43	3.92/5
Fall 2011	MATH 141	Calculus I	4	63	4.05/5
	MATH 708	Foundations of Comput. Math. I	3	8	4.43/5
Spring 2010	MATH 141	Calculus I	4	55	4.61/5
Fall 2009	MATH 526	Numerical Linear Algebra	4	20	4.31/5
Spring 2009	MATH 242	Elementary Differential Equations	3	36	4.74/5
Fall 2008	MATH 526	Numerical Linear Algebra	4	17	4.27/5
Spring 2008	MATH 706	Numerical Linear Algebra	3	6	4.78/5
Fall 2007	MATH 520	Ordinary Differential Equations	3	24	4.54/5
Spring 2007	MATH 727	Numerical Analysis II	3	4	4.00/4
Fall 2006	MATH 526	Numerical Linear Algebra	4	17	3.64/4
	MATH 726	Numerical Analysis I	3	7	4.00/4
Spring 2006	MATH 141	Calculus I	4	57	2.11/4
	MATH 526	Numerical Linear Algebra	4	19	3.38/4

Fall 2005	MATH 728J	Scientific Computing I	3	5	4.00/4
Spring 2005	MATH 526	Numerical Linear Algebra	4	23	3.64/4
Fall 2004	MATH 241	Vector Calculus	3	46	3.21/4

Research Supervision Experiences

In addition to regular teaching, another important task of my duties at University of South Carolina is research supervision to graduate students, undergraduates, and postdocs. I have been the member of both graduate and undergraduate advisory committees in the department providing help to students in taking courses and to the department in recruiting new students and developing new courses.

I so far have finished supervising/co-supervising 9 Ph.D. students:

- Li Tian (Ph.D., Mathematics, 2009/8, Dissertation title: *"Error Estimates for Finite Element/Volume Approximations of Dissipative Partial Differential Equations on Surfaces"*)
- Xiao Xiao (Ph.D., Mathematics, 2012/5, Dissertation title: *"Covolume-upwind Finite Volume Approximations for Linear Elliptic Partial Differential Equations"*)
- Yu Cao (Ph.D., Computer Science and Engineering, 2013/8, co-supervisor, Dissertation title: *"3D Grain Segmentation In Superalloy Images Using Multichannel Edge- Weighted Centroidal Tessellation Based Methods"*)
- Youjie Zhou (Ph.D., Computer Science and Engineering, 2015/8, co-supervisor, Dissertation title: *"Propagated Image Segmentation Using Edge-Weighted Centroidal Voronoi Tessellation Based Methods"*)
- Chenfei Zhang (Ph.D., Mathematics, 2019/8, Dissertation title: *"Unconditionally Energy Stable Linear Schemes for a Two-Phase Diffuse Interface Model with Peng-Robinson Equation of State"*)
- Shuai Yuan (Ph.D., Mathematics, 2020/8, co-supervisor, Dissertation title: *"An Ensemble-based Projection Method and its Numerical Investigation"*)
- Zhenyao Wu (Ph.D., Computer Science and Engineering, 2022/8, co-supervisor, Dissertation title: *"Learning Depth from Images"*)
- Xinyi Wu (Ph.D., Computer Science and Engineering, 2022/8, co-supervisor, Dissertation title: *"Cross Domain Semantic Segmentation"*)
- Yuankai Teng (Ph.D., Mathematics, 2023/8, Dissertation title: *"Deep Learning Methods for Some Problems in Scientific Computing"*)

and 5 M.S. students:

- Wensong Wu (M.S., Mathematics, 2007/8, Thesis title: *"Adaptive Finite Volume Methods for Convection-Diffusion Equations with Mesh Optimization"*)
- Xiao Xiao (M.S., Mathematics, 2010/8, Thesis title: *"Over-relaxation Lloyd Method for Computing Centroidal Voronoi Tessellations"*)
- Jing Liu (M.S., Mathematics, 2010/8, Thesis title: *"Construction of Centroidal Voronoi Tessellations Using a Conjugate Gradient Method Based on Trust Regions"*)
- Rosalia Tatano (M.S., Mathematics, 2013/5, Thesis title: *"Study on Covolume-Upwind Finite Volume Approximations for Linear Parabolic Partial Differential Equations"*)
- Meshack Kiplagat (M.S., Mathematics, 2013/12, Thesis title: *"The Compact Implicit Integration Factor Scheme for the Solution of Allen-Cahn Equations"*)

In particular, Dr. Tian was one of the four winners of *USC Dean's Award for Excellence in Graduate Study* in 2009. I currently have three Ph.D. students, Yuwei Geng (co-supervisor), Anusooya Vasudevan and Shuting Wang, in Mathematics Department. In addition, I have been a member of the Doctoral Dissertation Committees of 18 Ph.D. students since 2004.

I also have been actively involved in postdoctoral training. Many postdocs have been supported by my research grants at University of South Carolina and worked on diverse projects, including:

- Dr. Huai Zhang (2008/9–2010/3)
- Dr. Tong Zhang (2013/8–2015/8)
- Dr. Thi-Thao-Phuong Hoang (2017/1–2018/8)
- Dr. Huadong Gao (2018/3–2018/12)
- Dr. Xiao Li (2018/8–2019/8)
- Dr. Xucheng Meng (2018/9–2019/8)
- Dr. Anthony Gruber (2021/1–2022/8)
- Dr. Rihui Lan (2020/5–2023/12)

I also co-mentored three postdocs, Dr. Xiaoran Shi (2012/7–2013/7), Dr. Hao Tian (2013/7–2015/1) and Dr. Jianfang Lu (2016/7–2018/8) at BCSRC in China.

In addition, I have supported and hosted many visiting scholars for research collaboration at University of South Carolina, including Yang Li (2010/8–2011/2), Weidong Zhao (2013/9–2013/11), Wei Leng (2013/9–2013/12), Liyong Zhu (2014/3–2015/3), Jian Zhang (2015/10–2015/11), Hongwei Li (2016/9–2017/9), Chunbao Zhou (2017/1–2017/8), Xiaoping Zhang (2018/6–2018/8), Jianguo Huang (2018/8–2018/9), Kun Jiang (2019/9–2020/9), and Jingwei Li (2019/10–2020/10).

In 2006, I was the faculty advisor of USC teams for the Mathematical Contest in Modeling (MCM), consisting of undergraduate students. In that year, the USC team with William Kay and Raleigh Youmans as members won the MCM *Honorable Mention* award under my supervision.

Teaching Philosophy

I do believe that a good teacher is not only to teach effectively what students need to study, but also to motivate them to learn more. Mathematics can be really intimidating for some students, while others may find slow-paced classes boring, so there should be enough flexibility in the difficulty level allowing for a healthy balance. Additionally, mutual respect is also very important and is a good way to maintain the teaching quality and students' interests to the courses. I believe students tend to learn better and are more open to new ideas when they are surrounded by a friendly, learner-centered environment, where they feel apt and their opinions are well valued. It is important that they feel comfortable to ask questions and are encouraged by the instructor to do so.

Diversity is an important aspect I have paid special attention to, especially in research supervision. I have mentored quite a few female graduate students and postdocs, for example, Dr. Thi-Thao-Phuong Hoang is now a tenure-track faculty in Auburn university who received the NSF Career award in 2021, Dr. Xiao Xiao and Dr. Xinyi Wu are researchers in IT industry, and Dr. Rosalia Tatano is a lead scientist of Addfor Industriale in Italy. In addition, Mr. Meshack Kiplagat is African American student who successfully obtained his Masters degree in Mathematics under my supervision.

To summarize, I believe that my teaching and research experience and potential have prepared me to take on the challenges of being an outstanding researcher as well as an excellent teacher. I really enjoyed communicating with students and watering their thirst for knowledge. I am willing to devote myself to mathematics teaching since I find teaching to be a very rewarding experience.